

Real-World Reference Research for a Brazilian Foundation Shade-Matching Website

TL;DR

- **Findation.com is the closest reference product:** a crowdsourced, cross-brand shade-translation database ("powered by over 159 million users, over 11,000 products and 106,000 shades," per its homepage) built by Adore Beauty founder Kate Morris; your project should replicate its "enter a shade you already own → get matches in other brands" model while adding a Brazilian "where to buy" layer and an undertone quiz — features competitors handle either with questionnaires (Fenty, Sephora Color IQ) or AI photo analysis (L'Oréal Match My Shade, Perfect Corp/YouCam).
- **A defensible data model exists:** undertone (quente/frio/neutro/oliva), finish (matte/acetinado/luminoso/natural), coverage (leve/média/alta), texture (líquida/cremosa/mousse/bastão/pó/cushion) and Fitzpatrick-style depth are the industry-standard attributes; shade matching itself is best justified with the CIELAB color space + Delta-E (CIEDE2000) distance, the documented industry standard used by Sephora/Pantone, ShadeVault, and academic/patent literature.
- **Brazilian retail reality:** Beleza na Web (Grupo Boticário) and Época Cosméticos (Magalu) lead beauty e-commerce (both above 8 million unique visitors per Comscore, March 2023), with Sephora Brasil (LVMH) close behind; none of them exposes a "subtom" (undertone) filter, so an undertone-keyed "where to buy in Brazil" feature — keyed on price, installments (parcelamento), shade options and store — is both feasible and a genuine differentiator.

Key Findings

1. **Findation is a crowdsourced cross-brand translation engine, not a photo scanner.** Users enter one or more foundation/concealer shades they already wear; a proprietary algorithm returns matching shades in other brands, each with a percentage confidence score. It is embedded on retailer product pages (e.g., Adore Beauty) via a "Find Your Shade"/"Find My Match" button — a B2B2C model.
2. **Undertone determination is standardized around 4–6 self-assessment tests** (vein, jewelry, sun reaction, white/cream fabric, lip color), typically presented as a 5–6 question quiz mapping to warm/cool/neutral (plus olive). Brazilian sources use the same framework in Portuguese: quente/frio/neutro/oliva.
3. **Finish and formula vocabularies are consistent industry-wide** and map cleanly to Portuguese equivalents used by Brazilian retailers.
4. **CIELAB + Delta-E (CIEDE2000) is the documented gold standard** for perceptual color matching, used commercially and in peer-reviewed and patent literature, with published interpretation thresholds.

5. **Sephora/Bazaarvoice-style review systems attach reviewer demographics** (skin tone, skin type, undertone, eye/hair color, age) as structured, filterable attributes.
6. **Brazilian e-commerce data models** center on marca, acabamentoo, cobertura, textura, tipo de pele, price ranges, parcelamento and shade "opções"/variations — but not undertone.

Details

1. Direct competitors and similar tools

Findation.com — The single most relevant reference. Founder Kate Morris describes it as "a crowdsourced online database of matching foundation shades — all you have to do is put in the brands and shades that match your skin, and Findation can recommend your best match in other brands." Key facts:

- **Scale:** The homepage states verbatim that it is "powered by over 159 million users, over 11,000 products and 106,000 shades." An earlier third-party writeup (The Beauty Foodie) cited a much smaller figure — "Findation uses data from over 553,000 foundation users worldwide and a sophisticated proprietary algorithm to match shades between brands" — illustrating that the user count is a growing, self-reported marketing number.
- **Method:** Enter ≥ 2 -3 foundations you already own that genuinely match; more reference shades = better accuracy. Results come with a percentage match rating. It matches across liquid, mineral, loose and pressed foundations. (Findation)
- **UX guardrails** (directly from the site): only enter shades you'd buy again; check packaging rather than guess; enter at least one full-coverage foundation (sheer/tinted-moisturizer-only inputs degrade the result); for seasonal users, enter the shade that works year-round.
- **Business model / origin:** Built by Kate Morris, who also founded Australian beauty e-commerce site Adore Beauty in 1999 (from a garage with an A\$12,000 loan). Per SmartCompany, she launched Findation after "noting numerous online forum posts comparing the thousands of different shades of foundation." It is embedded on retailer complexion product pages ("Find Your Shade" button) as a conversion/affiliate tool rather than a direct storefront.

Sephora + Pantone Color IQ — Launched July 2012 in collaboration with the Pantone Color Institute. An in-store handheld spectrophotometer (Pantone CAPSURE) scans skin at three points (forehead, cheek, jaw), takes 27 color-accurate images, (Trend Hunter) assigns an official **Pantone SkinTone number** (e.g., "3Y09," where "Y" = yellow/undertone and trailing digits = depth), then matches against Sephora's Universal SkinTone Library of 1,000+ foundation SKUs on an iPad. (PR Newswire) Per Digiday, "since its launch, Sephora stores have generated 14 million Color IQ matches and the company has also created a spinoff, the Lip IQ." A web version lets users enter a shade they already wear to get matches — conceptually identical to Findation.

L'Oréal — True Match, Match My Shade, ModiFace — Multiple relevant references:

- **True Match** organizes 45-47 shades into cool/neutral/warm, labeled on the bottle with C / N / W codes plus depth numbers (e.g., C0.5, N5.5) — a clean example of an undertone-coded shade taxonomy.
- **Match My Shade** is an AI skin-scanner: the user records a 360° selfie video, inputs skin type + desired coverage + finish, and gets recommendations. L'Oréal's own service page states it works "with a skin-matching accuracy of 95%" and "compares your unique skin with a catalogue of over 45,000 skin tones."
- **ModiFace** (acquired by L'Oréal in 2018) powers AR virtual try-on and shade recommendation. Per Harvard Business School's Digital Initiative case, ModiFace enables "precise measurement of 68 unique facial parameters" and, in an early trial "scanning a sample of 1.6 million photos across 364,012 users," "accurately detected skin tone with 98.3% accuracy."

Perfect Corp / YouCam Makeup — The dominant B2B beauty-tech vendor. Per Perfect Corp's own press materials, "its YouCam suite of apps has been downloaded over 1.1 billion times globally" (the "over 1 billion" milestone was first announced in March 2022), with a network of over 800 global brand partners. Its AI Smart Shade Finder detects skin tone and undertone from a photo and recommends foundation with AR try-on; Taiwan Excellence's listing states the tool can "detect and match customer skin tone with up to 89,969 shades available." The Make Up For Ever partnership (Business Wire, April 20, 2020) used "AI smart detection based on over 200,000 learning samples across all skin groups." This is the reference for the "photo-based" alternative to Findation's "known-shade" approach.

Fenty Beauty Shade Finder — Highly relevant for quiz design. The original version was a **6-question interactive quiz** (jewelry preference, sun reaction, etc.). Flow: (1) "Have you found your shade?" (2) choose shade family (Light 100s / Medium 200s / Tan 300s / Deep 400s), (3) select a model who resembles you, (4) select undertone from **cool / neutral / warm / warm golden**, with an embedded undertone sub-quiz for the unsure. It then recommends Pro Filt'r foundation shades plus matched concealer/powder/etc. Fenty's shade numbering encodes both depth and undertone (e.g., 110; 225 "neutral olive"; 290 "warm olive"; 470 "cool neutral"). A newer version uses a selfie + AI. Fenty's sun-reaction heuristic: burns easily → cool; tans easily → warm; burns then tans → neutral.

Other tools: Maybelline Foundation Finder; **Ulta Shade Finder** (select the brand/formula/shade you wear on any complexion product page → get a match — a direct Findation-style feature); Match My Makeup (science-based); MakeupMatcher "Foundation Finder" (consultant app, photo → Dynamic Color Wheel, \$4.99/mo); and independent CIEDE2000 matchers ShadeVault and foundation-shade-finder.com. A 2026 RevenueBerry test named the core hard problems: cameras warm-correct under indoor LED (pushing cool skin to "neutral/warm"), training-data bias toward light-to-medium depth, and undertone-vs-depth confusion.

Brazilian/Latin American tools: No dedicated Brazilian cross-brand shade-matcher equivalent to Findation was found — a genuine market gap. Brazilian brands provide educational content (Maybelline Brasil, Eudora, Avon Brasil, and the Sephora Brasil blog all publish subtom guides) and some AR mirrors (L'Oréal, Boticário, Sephora), but no known "enter your shade → find matches across Brazilian-available brands" product.

2. Undertone (subtom) determination methods

The consumer-facing consensus, consistent across L'Oréal, Maybelline, Jane Iredale, Fenty and Brazilian retailers:

- **Vein test:** Look at inner-wrist veins in natural light. Blue/purple → cool; green → warm; mixed/hard to tell → neutral. (Beleza na Web) Widely presented but repeatedly flagged as unreliable alone (Maybelline BR explains the science: a yellow/warm skin filter makes veins read green). Fails on deep skin where veins aren't visible.
- **Jewelry test:** Gold flatters → warm; silver flatters → cool; both → neutral. One color-consultant source calls it "the test that professional color consultants rely on most heavily," (HueCheck) though preference bias is a caveat.
- **Sun reaction (Fitzpatrick-style):** Burns easily/rarely tans → cool; tans easily/rarely burns → warm; burns then tans → neutral. This maps onto the **Fitzpatrick phototype scale (I–VI)**, developed by Thomas Fitzpatrick in 1975 based on burn/tan response: Type I always burns/never tans → Type VI never burns/deeply pigmented. (uspto) Fitzpatrick is a depth/UV-reactivity measure, distinct from undertone; combine both.
- **White vs. cream fabric test:** Pure white flatters → cool; cream/ivory flatters → warm; both → neutral.
- **Lip/eye color and "which colors suit you":** secondary signals.

Presentation in quizzes: Typically **5–6 questions**, each a test above, tallied to warm/cool/neutral (olive treated as a special neutral-with-green case). The best-practice message stated repeatedly: no single test is definitive; use multiple in natural daylight on bare skin and go with the majority. This directly justifies a multi-question weighted-scoring quiz rather than a single-question result.

Brazilian terminology (important for localization): subtom **quente** (warm — dourado/amarelado/pêssego), **frio** (cool — rosado/azulado), **neutro** (balanced), and **oliva** (olive — a green/greyish base treated as a distinct fourth category, widely discussed in Brazil as under-served by commercial shade ranges). Brazilian tom (surface depth: claro/médio/escuro/retinto) is distinguished from subtom (stable underlying temperature). Avon Brasil notes Color Trend bases coded "N" (neutros) and "F" (frios); MAC's "C"/"N"/"W" and L'Oréal's C/N/W are referenced by Beleza na Web.

3. Finish and formula terminology

Finishes (consistent across Honest, Escentual, e.l.f., Beautyblender, BuyMeBeauty):

- **Matte:** no shine, dries flat, minimizes pores; best for oily/combination skin.

- **Semi-matte / soft-matte / velvet:** a newer velvety category that controls oil without over-drying; good for combination skin.
- **Satin (acetinado):** the middle ground between matte and dewy; "the most common foundation finish"; suits all skin types; feels slightly heavier.
- **Dewy / radiant / glow (luminoso):** reflective, luminous, uses oils/humectants; best for dry/mature skin.
- **Natural / sheer:** light, skin-like (often BB/CC creams, skin tints).
- Note: finish is independent of coverage — a foundation can be full-coverage matte or full-coverage dewy.

Formula/texture types (bareMinerals, Well People, Watsons, Nanshy, KBL Cosmetics):

- **Liquid (líquida):** most versatile; water-, oil-, or silicone-based; all finishes/coverage; (Ana Coutinho) includes BB/CC creams and skin tints as sub-types.
- **Fluid:** a lighter, thinner liquid (marketing term, e.g., MAC Studio Fix Fluid, Fenty Fluid Flex).
- **Cream (cremosa):** thicker, hydrating, medium-full coverage; (Sindebella) pots or sticks; dry/mature skin. (Sephora)
- **Mousse/whipped:** whipped, airy, doesn't settle into fine lines; mature skin. (Blogger)
- **Stick (bastão):** cream in stick form; portable, buildable/targeted coverage, quick touch-ups. (bareMinerals) (Sephora)
- **Powder (pó):** pressed or loose; lightweight, oil-absorbing, matte; (Colors Queen) oily skin, humidity. (Sephora)
- **Cushion:** Korean origin; liquid foundation in a sponge-filled compact with a puff; (Nanshy) dewy, light-medium coverage, (Sindebella) often with SPF/skincare.
- **Cream-to-powder / cream-to-matte:** applies creamy, sets to powder. (Nanshy)

Retailer categorization: Retailers filter by these as facets. Cosmetics Business notes that even Sephora and Douglas often *don't* consistently offer skin-tone filters when browsing foundation because retroactively tagging thousands of SKUs is a heavy lift — meaning granular finish/coverage/undertone metadata is a competitive differentiator, and building it in from the start is an architectural advantage.

4. Where Brazilians buy makeup online (data model for "where to buy in Brazil")

Market leaders and structure:

- **Beleza na Web** (belezanaweb.com.br) — pioneer (founded 2008 by Alexandre Serodio); acquired by **Grupo Boticário** in 2019; (Ligausp) 10,000+ brands, 80,000+ products; added a marketplace in 2020 and a pro channel (Beleza na Web Pro) in 2021. Won E-Commerce Brasil's "most admired" cosmetics/perfumery award in 2020.
- **Época Cosméticos** (epocacosmeticos.com.br) — Brazil's first beauty e-commerce (1996); acquired by **Magazine Luiza (Magalu)** in 2013. (Ligausp)

- **Sephora Brasil** (sephora.com.br) — descends from Sack's (2000, "maior e-commerce de cosméticos de luxo da América Latina"), acquired by **LVMH in 2010**. (Diadebeaute) (Ligausp)
- Others: O Boticário, Eudora (Grupo Boticário), Adcos (dermocosméticos), Amazon Brasil, Mercado Livre.
- **Traffic:** Comscore (March 2023, via E-Commerce Brasil) found "Beleza na Web e Época Cosméticos liderando o ranking de retailers com uma audiência superior a oito milhões de visitantes únicos" — with Sephora, Lojas Rede and Ikesaki rounding out the top five, and beauty/cosmetics categories drawing 23.6M and 28.6M unique visitors. Cosmetics is consistently among the top categories in Brazilian e-commerce.

Product-page / filter attributes actually used (from direct inspection):

- **Época Cosméticos** (VTEX platform): filters for **Marca**, **Textura** (Compacta, Cremosa, Mousse, Bastão, Líquida, Tradicional), **Cobertura** (e.g., Alta), **Oil-Free**, **Vegano**, **Cruelty Free**, and **Faixa de preço** (confirmed brackets: "Até R\$ 69," "De R\$ 70 a R\$ 99," "De R\$ 100 a R\$ 169," "Acima de R\$ 170"). Sort: Mais vendidos, Menor/Maior preço, A-Z, Data de lançamento, Melhor desconto. (Época Cosméticos) Product pages show de/por price, **parcelamento** (e.g., "7x de R\$ 30,17"), site-wide "Frete Grátis em até 10x," and description fields Cobertura/Acabamento/Tipo de pele/Modo de Usar.
- **Sephora Brasil** (Salesforce Commerce Cloud): **Marca** filter; sort Mais Vendidos/Menores Preços/Maiores Preços/Novidades/Maior desconto; (sephora) each product shows a shade-option count ("(50 opções)," "(68 opções)") indicating a shade grid, inline price + parcelamento ("R\$ 345,00 ou 10x de R\$ 34,50"), (sephora) discount badges, and "SÓ NA SEPHORA"/"NOVIDADE" tags. Shade codes appear in names (NC15, 1N, 106N). Category vocabulary uses acabamentoo (matte/luminoso/acetinado/natural), cobertura (leve/média/alta), tipo de pele, and subtom (quente/frio/neutro) — though subtom is editorial, not a clickable PLP facet.
- **Beleza na Web:** filters via URL params **acabamento** (matte, luminoso, glow, natural), **textura** (líquida, bastão, cremosa, compacta, mousse), **tipo-de-pele**, plus **Vegano/Cruelty Free/Outlet** badges. Dual pricing: "de R\$ 73,39" Y pagando no pix ou R\$ 73,39), (Beleza na Web) shade count ("+ 67 opções"), star ratings ("nota 4.7 de 5"), frete grátis.
- **Mercado Livre:** shade handled as a **variação** on the attribute "**cor**", unifying shades under one anúncio (Conecta Ads) (variation fields: preço, título, EAN, atributo diferenciador such as cor, estoque). Cards show price, star rating + review count, frete grátis, units sold, and seller ("vendido por"). Parcelamento depends on listing type — an Anúncio Premium offers up to 12x sem juros via Mercado Pago; a Clássico does not.
- **Amazon Brasil:** shade as a variation/attribute ("Cor 13" in the title; parent/child ASINs); parcelamento up to 12x (com or sem juros via NuPay); (Amazon) shows "vendido por" (seller) + "enviado por" (fulfillment), "Enviado pela Amazon" (FBA), (Amazon) and "A Escolha da Amazon" badge.

Data-model implication: a "where to buy" record should hold: store name, product URL, shade/variation identifier, current price, pix vs. cartão price, discount %, parcelamento terms, frete-grátis flag, availability/estoque, and (on marketplaces) seller. Critically, **none of these retailers exposes a "subtom" (undertone) filter** — so an undertone-driven catalog is your differentiator.

5. Review systems with reviewer skin-tone/type disclosure

- **Sephora:** reviews attach structured reviewer attributes — **skin tone, skin type, eye color, hair color, age range** — alongside rating and text, powering filter chips so shoppers can find reviewers like themselves. (Confirmed via the Sephora review data structure surfaced by third-party datasets that enumerate exactly these demographic fields per review.)
- **Bazaarvoice** (the dominant ratings-and-reviews platform behind many beauty retailers): its "Dynamic filters" let a brand feature the most important attributes on a product, "from the different colors of a product to the skin type of a reviewer." It also **syndicates** a review collected on a brand site to retailer product pages across its network (Walmart, Target, Walgreens, etc.) — relevant if your project ever aggregates reviews.
- **Best-practice attribute set** (per a 2026 beauty AI-commerce playbook): beauty-specific structured signals matter more than generic ones — skin type compatibility (oily/dry/combo/sensitive), undertone, shade, finish, coverage level, ingredient flags (paraben-free, fragrance-free, vegan, cruelty-free), and formulation type. Structuring shade/finish data and capturing reviewer demographics is explicitly recommended.
- **Structure:** implement reviewer demographics as enumerated, filterable tags (chips) — e.g., a review record referencing user attributes {tom, subtom, tipo_de_pele, idade} plus rating and body — enabling "ver avaliações de quem tem meu subtom."

6. UX / data-model guidance for shade-matching engines

- **Color space:** The documented industry standard is *CIELAB (CIE Lab)***, defined by the CIE in 1976 so that Euclidean distance approximates human-perceived color difference (unlike RGB). L = lightness (0-100), a^* = green-red, b^* = blue-yellow.
- **Distance metric: Delta-E**, with **CIEDE2000** the current perceptual standard. ShadeVault states it "uses the CIEDE2000 Delta-E algorithm — the industry standard for perceptual color difference," converting swatches to Lab^* and computing distance, with published thresholds: **"A Delta-E score of 0-2 means the colors are nearly identical (imperceptible difference). Scores of 2-5 indicate a great match visible only on close inspection. Scores of 5-10 show similar colors with noticeable differences. Above 10, the colors are clearly different."** foundation-shade-finder.com measures skin against 6,752 real shades using CIEDE2000. The simpler ΔE (CIE76) formula is $\sqrt{((L_2 - L_1)^2 + (a_2 - a_1)^2 + (b_2 - b_1)^2)}$. ([arxiv](#))
- **Academic/patent grounding:** A 2026 *Color Research & Application* (Wiley) paper classifies skin into six clusters via CIELAB + k-means (deltaE76 for cluster

assignment), explicitly citing foundation-shade development as an application. An NCIRL thesis ("Foundation Makeup Shade Recommendation using Computer Vision") compares RGB/HSV/YCrCb color spaces and Delta-E to match a face image to a dataset of ~700 foundation shades. A US patent ("Diverse cosmetic and skin care product matching system") documents a production approach: the scan device produces one color space (Lab) while products are indexed in another (LCH), matching across Lab/LCH/YUV/HCL/CIELAB/RGB/TSL.

- **Architecture takeaways for your project:**

- Store each shade as measured/estimated Lab* coordinates plus categorical metadata (brand, formula, finish, coverage, undertone label, depth).
- For cross-brand matching (the Findation model), translate a known shade to Lab* then rank candidate shades by CIEDE2000 distance; surface a percentage/ ΔE confidence.
- The "known shade $\rightarrow$ matches" (Findation/Ulta) approach is more robust than photo analysis for a student project because it avoids the camera-calibration, lighting and undertone/depth-confusion problems documented in shade-finder app testing. A photo pipeline can be a later phase.
- Depth is easier to match than undertone; deep shades are under-represented (one matcher reports 64% of shades sit in the lighter half of the range), so a cross-brand shortlist genuinely helps deeper skin tones.

Recommendations

Phase 1 (MVP — core requirement): Build the Findation-style engine: a normalized catalog of foundations (brand, product line, shade name/code) each tagged with undertone (quente/frio/neutro/oliva), depth, finish (matte/acetinado/luminoso/natural), coverage (leve/média/alta) and texture (líquida/cremosa/mousse/bastão/pó/cushion). Let users enter 1-3 shades they own and return ranked cross-brand matches. If you can source or estimate Lab* per shade, rank by CIEDE2000 and show a % confidence with the ΔE thresholds above (0-2 excellent, 2-5 good). If not, fall back to categorical matching on undertone + depth + finish.

Phase 2 (undertone quiz): Implement a 5-6 question weighted quiz (vein, jewelry, sun reaction, white/cream fabric, lip color), scoring to quente/frio/neutro/oliva, and explicitly tell users no single test is definitive — mirroring Fenty/Sephora/L'Oréal and Brazilian retailer guides. Use it to pre-filter the catalog.

Phase 3 (where to buy in Brazil): For each matched shade, store per-retailer offers with: store, URL, shade/variation ID, price, pix vs. cartão price, discount %, parcelamento, frete grátis, availability, seller. Prioritize Beleza na Web, Época Cosméticos, Sephora Brasil, Amazon Brasil and Mercado Livre. Because none of them filters by subtom, your undertone-keyed catalog is the wedge.

Phase 4 (reviews with disclosure): Add reviews where each reviewer discloses tom, subtom, tipo de pele (and optionally age); expose these as filter chips ("ver avaliações de quem tem meu subtom"). Model demographics as enumerated attributes on the user, referenced by the review, following the Sephora/Bazaarvoice pattern.

Benchmarks that change the plan: If you can obtain spectrophotometer/Lab* data for a meaningful share of shades, prioritize the color-space engine (Phase 1 color path) over categorical matching. If you cannot reliably populate per-retailer pricing (no API/scraping access), make "where to buy" a simple outbound-link directory rather than a live price aggregator. If you add a photo pipeline, budget for white-balance correction and test across depths, since documented failure modes cluster on deep skin and indoor LED lighting.

Caveats

- **Findation's user/shade counts** (159M users / 11,000 products / 106,000 shades on its homepage; an older third-party figure of 553,000 users) are self-reported marketing numbers, not audited figures.
- **AI accuracy and scale claims** — L'Oréal Match My Shade "95% accuracy" / "45,000+ skin tones," ModiFace "98.3% accuracy" on 1.6M photos / "68 facial parameters," Perfect Corp "1.1B+ downloads" / "89,969 shades" / "200,000 learning samples," Sephora "14 million Color IQ matches" — are vendor or press figures, not independently verified.
- **Brazilian retailer filters:** facet rails are JavaScript-rendered; the confirmed attributes above are solid, but the *complete* filter inventory (especially whether acabamentoo/cobertura appear as clickable facets on every base PLP) could not be exhaustively enumerated, and no retailer's undertone filter was found. Sephora reviewer skin-tone/skin-type tagging is confirmed for Sephora US but was not verifiable on the sephora.com.br site.
- **Traffic/market-share** figures are from Comscore via trade press (2023) and third-party analytics; AI-referral shares cited in some trade sources are estimates, not official.
- The **olive/oliva undertone** is inconsistently handled across the industry (sometimes a warm-leaning neutral, sometimes its own category) — decide on one canonical definition in your data model.
- **Fitzpatrick is a UV-reactivity/depth scale, not an undertone scale;** use it for depth self-assessment, not to derive warm/cool.